SunGeun **AN**

ssunggun2@snu.ac.kr | ssunggun2.github.io | linkedin.com/in/sunggeun-an | github.com/ssunggun2

RESEARCH INTEREST

- Whether multimodal language models reason from the *evidence in front of them* or from priors learned during pretraining.
- Benchmark and metric design that separates *perception* failures from *reasoning* failures.
- Evaluation methodology for language models, including pass@k diagnostics and human-baseline studies.
- **Keywords**: Multimodal LLMs, physical reasoning, evaluation and benchmark design, retrieval-augmented generation, vision-language models.

EDUCATION

03/2024 – 08/2026

Seoul National University
M.S. in Data Science
Graduate School of Data Science (GSDS), SKI-ML Lab
Thesis: UNREAL — Uncovering Physics Reasoning Limits in MLLMs

2012 – 2021

Seoul National University
B.S. in Geography Education
College of Education, Department of Geography Education

EXPERIENCE

03/2024 – 08/2026

M.S. Student, Research Assistant
Seoul National University, SKI-ML Lab, Seoul, South Korea

– Master's thesis (UNREAL): built a benchmark probing whether multimodal LLMs reason from the video in front of them or fall back on pretrained physical priors, synthesizing 1,528 videos in Kubric under 15 categories of deliberate physics violations with staged QA
– Proposed Reality Bias Rate and World-Model Inconsistency Rate as evaluation metrics and validated them against a 13-annotator human study
– Built PhysAgent, an agent pipeline that delegates perception to vision specialists (Depth-Anything-V2, RAFT, SAM2) so the MLLM handles reasoning only
– Industry collaboration with LG (2026): led evaluation and dataset research for extending a Korean LLM (EXAONE-3.5-7.8B-Instruct) beyond finance QA into long-tail expert domains, designing a pass@k diagnostic framework (N=128) that separates latent from trained capability and grades benchmarks into difficulty tiers by the pass@1 to pass@128 gap

12/2021 – 12/2023

ML/NLP Engineer
Aimpact, Seoul, South Korea

– Built and trained Korean language models for address and message extraction
– Designed and deployed NLP APIs using FastAPI, serving real-time inference
– Implemented OCR pipelines integrating Naver Clova and GCP Vision APIs
– Migrated backend from Flask to FastAPI with async processing for 3x throughput
– Integrated Redis caching and AWS SQS/Lambda for scalable async workflows
– Deployed and managed services on AWS ElasticBeanstalk

PUBLICATIONS

1. **Sieve and Sage: Efficient Distraction Filtering for Reliable RALM Abstention**.
   In *Findings of EMNLP*, 2026.
   Jongbin Won, **Sung Geun An**, Jay-Yoon Lee.

SERVICE AND ACTIVITIES

2024 – 2026	**Lab manager, SKI-ML Lab, Seoul National University** Operated the shared GPU cluster and LLM API gateway; ran the weekly seminar and onboarding program.
03/2026 – 07/2026	**Organizing committee, SNU AI Challenge 2026** Designed the Kaggle task and evaluation metric; scored final reports against a rubric.

SKILLS

Machine Learning & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
NLP & Multimodal	Large Language Models, Multimodal LLMs, Vision-Language Models, Retrieval-Augmented Generation, Named Entity Recognition, OCR
Evaluation & Benchmarking	Benchmark Design, Human Evaluation Study Design, pass@k Diagnostics, Prompting Strategy Analysis
Programming	Python, SQL
MLOps & Cloud	AWS (SQS, Lambda, ElasticBeanstalk), Docker, FastAPI, Redis
Languages	Korean (native speaker), English (professional working proficiency)